

BRC Documentation: Introduction to Hazel-HPC

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01. First steps in Hazel HPC

Set up an account

Request access from your supervisor and in the mean time read and understand [Acceptable Use Policy](#). Obtain your UnityID credentials and ensure you have VPN access if working off-campus.

Log in

1. **Web browser** (OnDemand): <https://servood.hpc.ncsu.edu>
2. **Terminal**: To connect to the shared HPC login nodes, open a terminal window and type

```
ssh [UnityID@login.hpc.ncsu.edu
```

where user_name is the Unity ID.

To allow Hazel to open a window on a local machine, e.g., for GUI applications, X11 forwarding must be enabled by adding the -X option (i.e. ssh -X user_name@login.hpc.ncsu.edu). On macOS, install XQuartz to use X11 forwarding. When using MobaXterm, X11 is enabled by default.

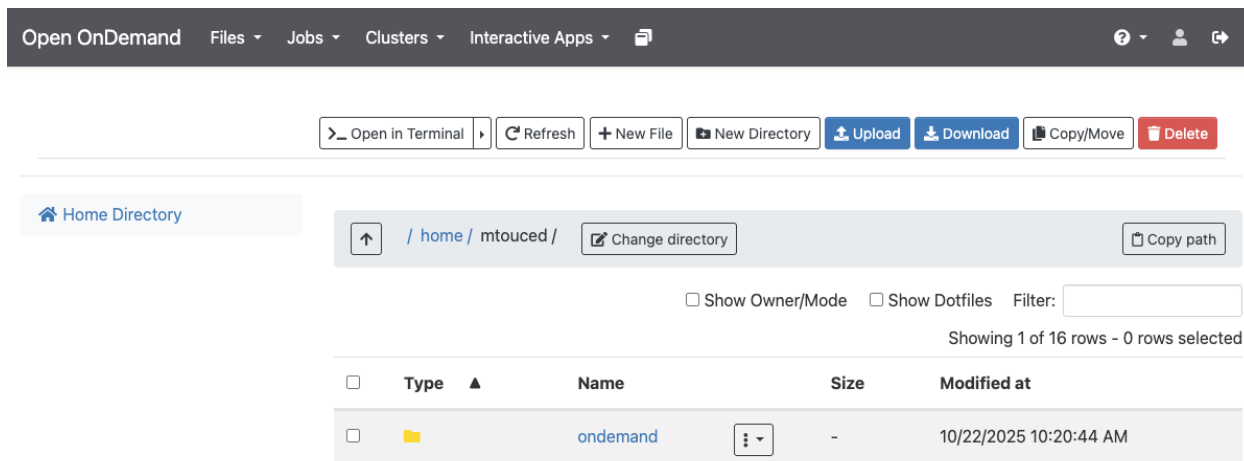
3. **VSCode**: <https://hpc.ncsu.edu/Documents/LoginUsingVSCode.php>
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Transfer files

OnDemand

Go to [OnDemand](#)

- Under **Files**, using the **Upload** or **Download** buttons to transfer. Make sure you navigate to the destination/source directory on cluster using the **Go To** button before transferring files.



SCP

For transferring a single file in one direction. Open a terminal on your local machine, and without login into the HPC do:

```
# Local to HPC
$ scp filename [UnityID@login.hpc.ncsu.edu:/rs1/researchers/[letter]/[name] local/path/to/stuff

# HPC to local
$ scp [UnityID@login.hpc.ncsu.edu:/rs1/researchers/[letter]/[name] local/path/to/stuff
```

SFTP

To transfer multiple files or in multiple directions (local to HPC and viceversa) without needing to re-authenticate, or to be able to navigate the remote file system. From the local machine initiate an SFTP session:

```
$ sftp [UnityID@login.hpc.ncsu.edu]
```

At the prompt do:

```
sftp> cd /path/to/stuff
sftp> put filename
sftp> get filename
```

Globus

Globus is a point to point file syncing platform. It is very useful for transferring large files or large numbers of small files, since it allows for background scheduled transfers. See our **Globus-ncsu tutorial** to learn more about Globus and how to use it with the NCSU Hazel HPC.

Useful commands

System and user info

- Check queues you have access to: `$ bqueues -u [UnityID]`
- Check the groups you belong to (first one is your default): `$ groups`
- To see what node you are in: `$ hostname` or `$ echo $HOSTNAME`

Storage

- Check size of the directory you are in: `$ du -sh .` > Important for checking pipeline output size
- Check size of files in a directory: `$ du -sh [pathtodirectory]/*`
- Check how much space is being used in your home directory: `$ quota`
- Check space allocations for your group: `$ quota -s -g group_name`
- Check files in reversed time order (most recently modified at the end): `$ ls -lrt`

Software and resource info

- Find available modules: `$ module avail`
 - Find path to executable: `$ wich [executablename]`
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Resources

- Getting started on Hazel: <https://hpc.ncsu.edu/Documents/GetStarted.php>
 - Cluster status: <https://hpc.ncsu.edu/Monitor/ClusterStatusMemory.php>
 - Support: `oit_hpc@help.ncsu.edu`
 - HPC-Henry2 moodle course on Reporter: <https://moodle-projects.wolfware.ncsu.edu/course/view.php?id=6857>
 - Youtube channel for OIT with tutorial videos: <https://www.youtube.com/@OITHPC>
 - To stay informed about Hazel-HPC news join the **HPC Announcements Google Group**
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Your First Week on Hazel HPC

Before You Start:

- ☐ Obtain your UnityID credentials and VPN access (if off-campus)
- ☐ Join your research group's allocation/project space
- ☐ Review this documentation and bookmark it for reference

Getting Connected:

- ☐ SSH into the login node: `ssh [UnityID]@hazel.ncsu.edu`
- ☐ Explore your home directory (`/home/[UnityID]`) - remember, only 1GB available!
- ☐ Locate your group's scratch space
- ☐ Find your group's RS1 storage